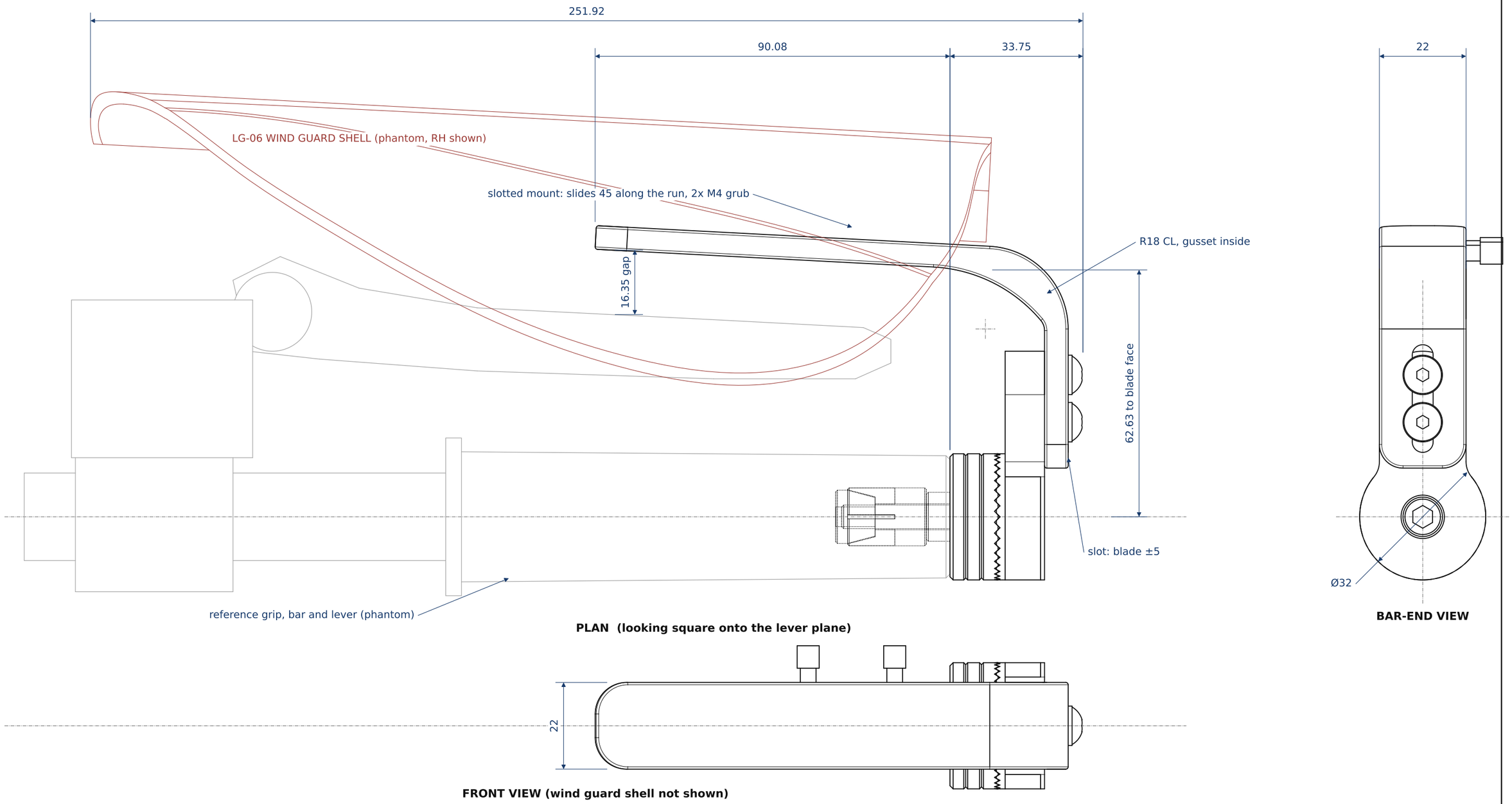


PLAN (looking square onto the lever plane)

BAR-END VIEW

FRONT VIEW (wind guard shell not shown)

NOTES

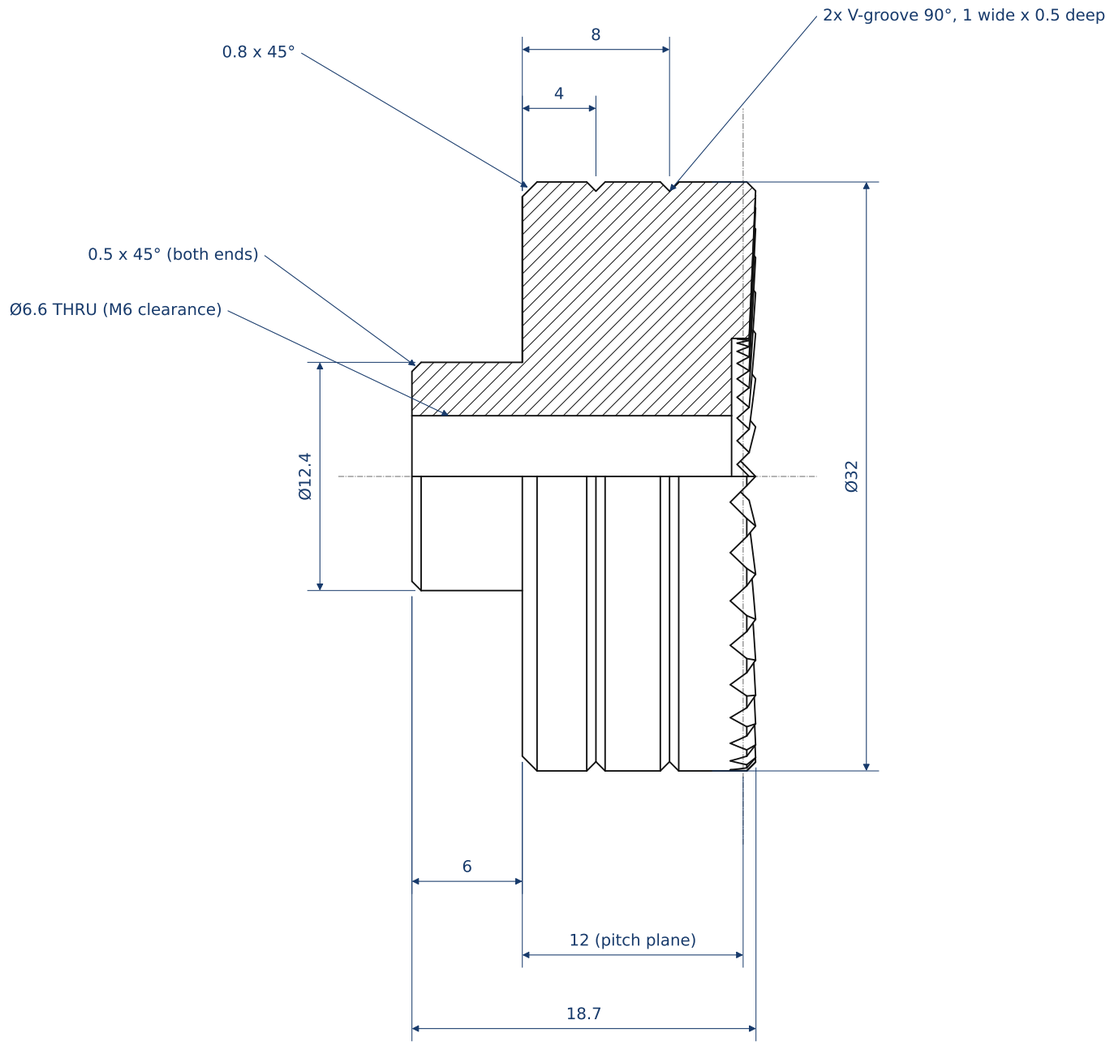
1. Drawing plane = lever plane. Turn the arm on its 36-tooth serration (10° steps) to put the blade on the lever line.
2. Clearances shown for the reference sport-bike lever; keep ≥ 10 mm all round, re-check at full lever travel.
3. Mass per side ≈ 123 g, 188 g with the wind guard.
4. Left side: same parts, arm turned over (they are symmetric about the lever plane), except the wind guard shell: LG-06 is made as a mirror-image pair.
5. Wind guard shell drawn as a phantom in the plan; its profile and fixing are on sheet 6.

RACING LEVER GUARD / UNIVERSAL BAR-END FIT

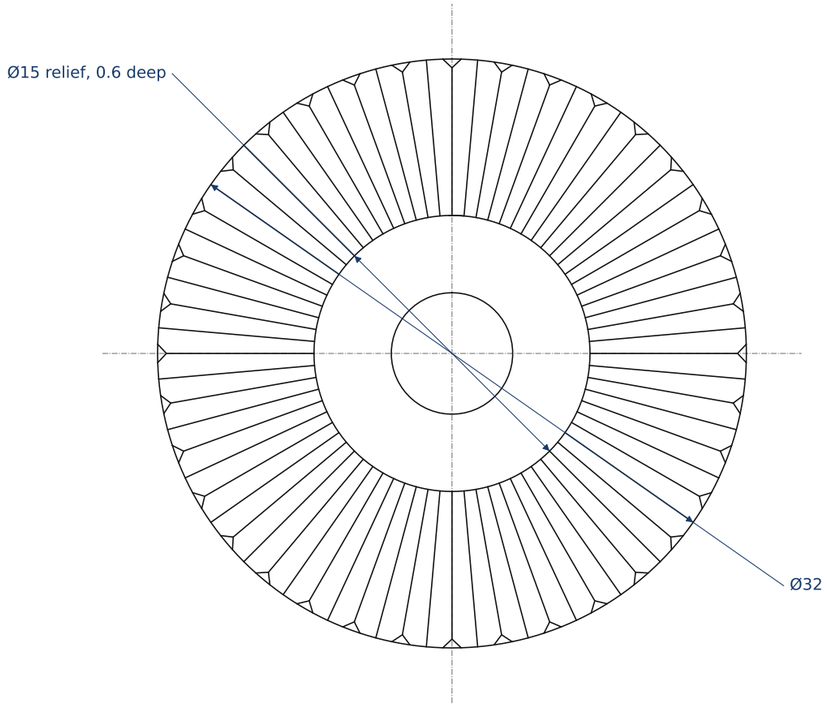
GENERAL ARRANGEMENT

LG-00

MATERIAL				FINISH	
see part sheets				see part sheets	
SCALE	UNITS	QTY / SIDE	GENERAL TOLERANCES	PROJECTION	SHEET
1 : 1	mm	1 set	ISO 2768-mK		1 / 6
SOURCE				DATE	BREAK SHARP EDGES
cad/lever_guard.py (CadQuery), drawings.py				2026-10-23	0.2 - 0.5



HALF SECTION A-A



VIEW ON SERRATION (from outboard)

NOTES

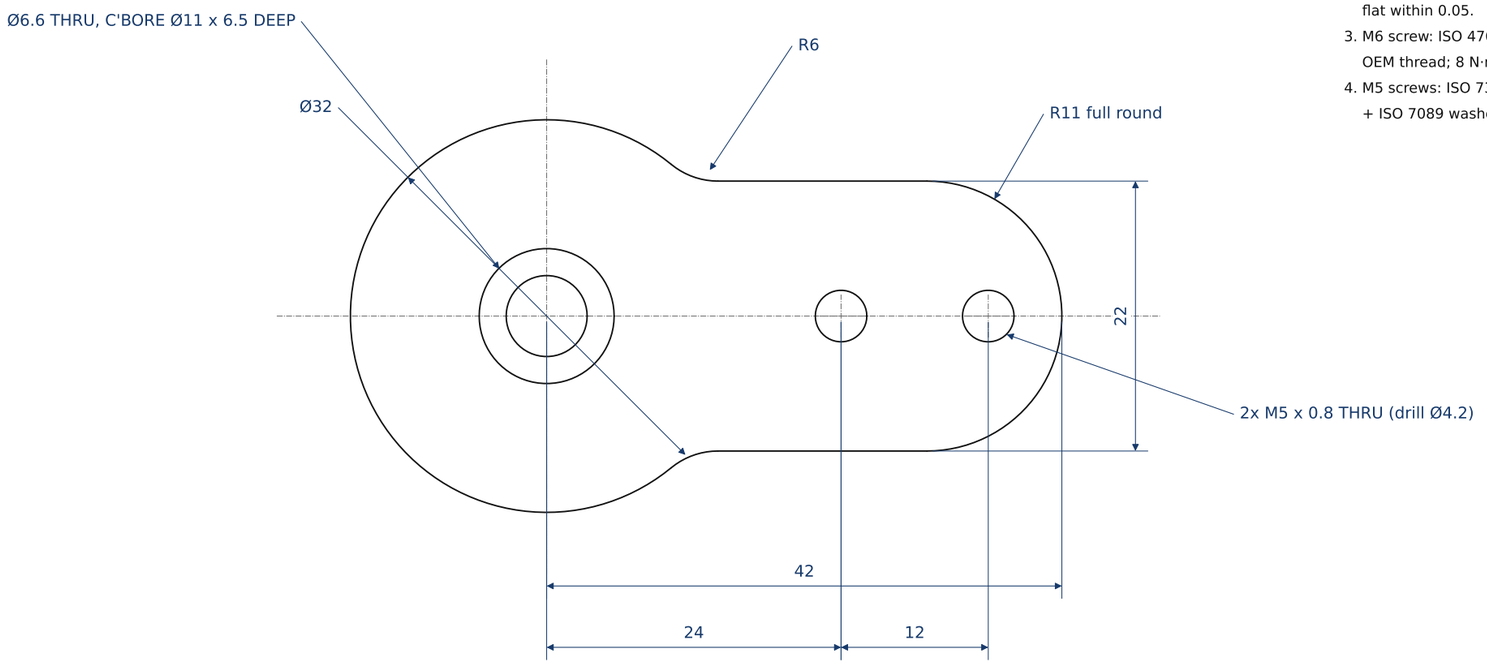
1. Hirth face serration: 36 teeth, 90° included, crests and roots radial through the axis, pitch plane at 12 from the bar-end face, toothed from Ø15 to Ø32: tooth height 1.4 at Ø32. Mates with LG-02 (offset half a tooth).
2. Variant LG-01b: same part without the spigot, for bars with a threaded OEM bar-end insert.
3. Spigot Ø12.4 centres in any bar with ID ≥ 12.5; the expander (LG-04/05) sits behind it.
4. Hard anodise after machining; mask nothing. Serration may be cut with a 90° V-cutter.

RACING LEVER GUARD / UNIVERSAL BAR-END FIT

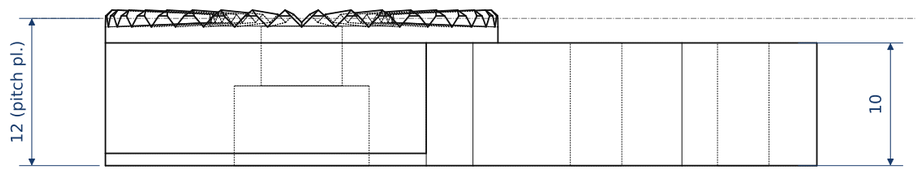
BAR-END ADAPTER

LG-01

MATERIAL			FINISH		
EN AW-6061-T6			Hard anodise, colour to order		
SCALE	UNITS	QTY / SIDE	GENERAL TOLERANCES	PROJECTION	SHEET
3 : 1	mm	1	ISO 2768-mK		2 / 6
SOURCE			DATE		BREAK SHARP EDGES
cad/lever_guard.py (CadQuery), drawings.py			2026-10-23		0.2 - 0.5



OUTBOARD FACE



PLAN (lever plane), serration toward the bar

NOTES

1. Hirth serration as LG-01: 36 teeth, 90°, pitch plane on the hub's inner face. Sits half a tooth out of phase with LG-01 so crests meet roots: 10° indexing.
2. Neck edges R1. Outboard face is the blade seat: flat within 0.05.
3. M6 screw: ISO 4762 M6 x 45 (expander) or to suit the OEM thread; 8 N·m, medium threadlocker.
4. M5 screws: ISO 7380 M5 x 14 (x 16 with LG-03b) + ISO 7089 washers, 4 N·m.

RACING LEVER GUARD / UNIVERSAL BAR-END FIT

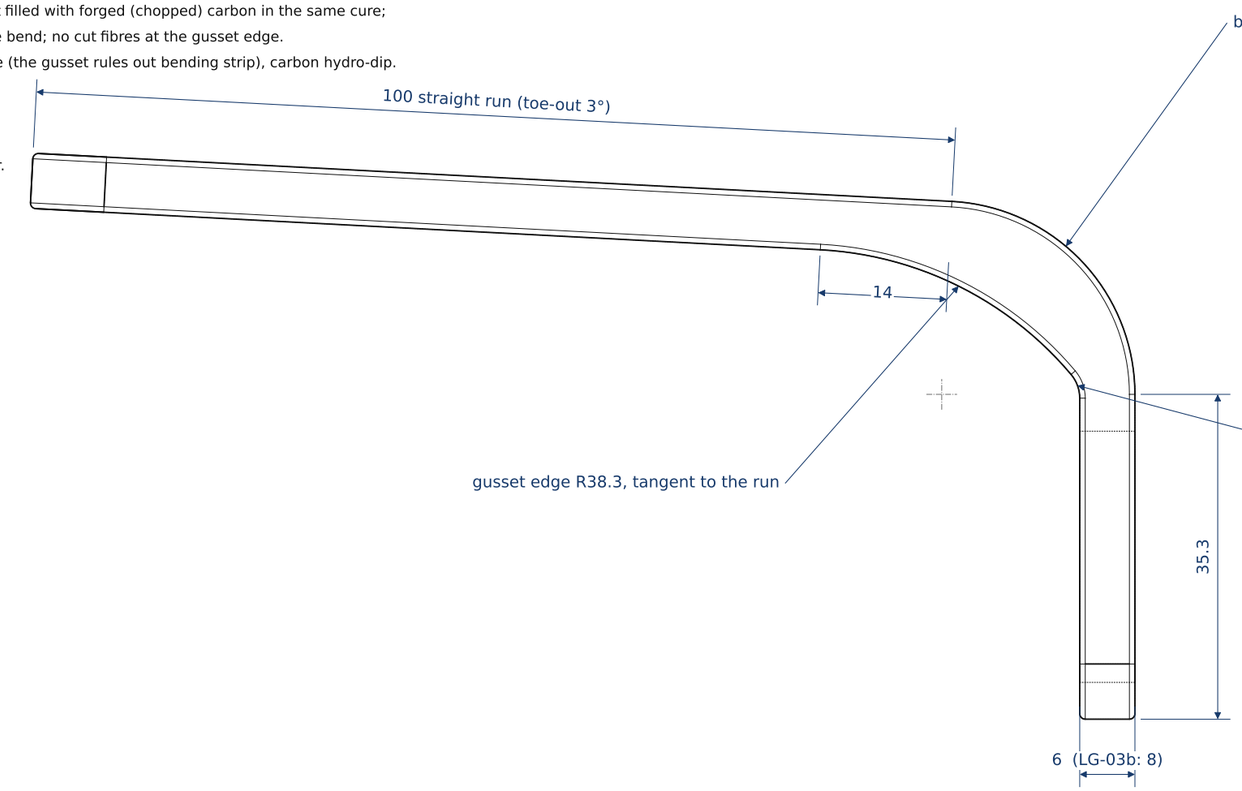
GUARD ARM

LG-02

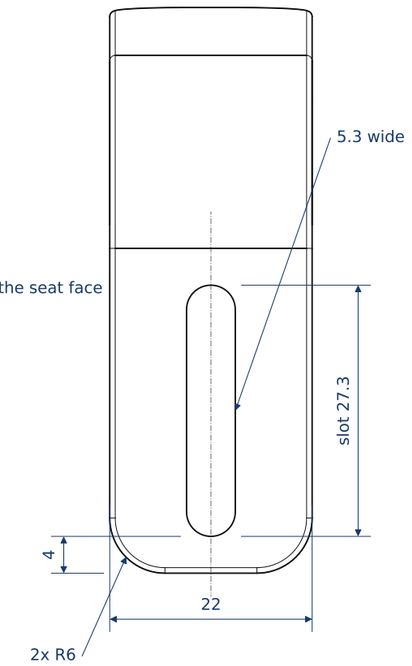
MATERIAL				FINISH	
EN AW-6061-T6				Hard anodise, colour to order	
SCALE	UNITS	QTY / SIDE	GENERAL TOLERANCES	PROJECTION	SHEET
2 : 1	mm	1	ISO 2768-mK		3 / 6
SOURCE				DATE	BREAK SHARP EDGES
cad/lever_guard.py (CadQuery), drawings.py				2026-10-23	0.2 - 0.5

NOTES

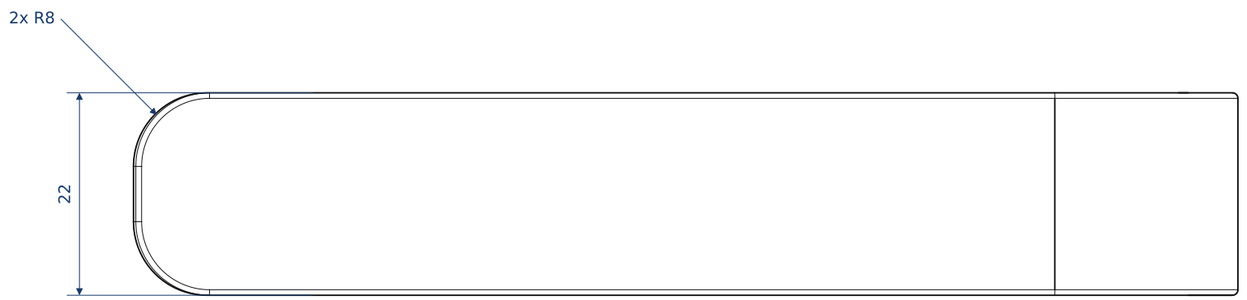
1. Centre-line length 162.6 mm: 35.3 straight + 87° bend on R18 + 100 straight.
2. Gusset inside the bend: section 6 -> 9.7 mm across the bend, about 2.6x the bending strength where a fall loads the blade.
   Its edge is tangent all along (blend into the seat face, arc, run-out on the run): no corner to start a crack.
3. LG-03 carbon: skin 6 ± 0.2 moulded over a male tool, gusset filled with forged (chopped) carbon in the same cure;
   CNC-trim the outline and slot. Lay plies continuous round the bend; no cut fibres at the gusset edge.
4. LG-03 aluminium option: machine from 6061-T6 25 mm plate (the gusset rules out bending strip), carbon hydro-dip.
5. LG-03b 3D print: same outline, 8 thick,
   PA12-CF / PETG-CF, lying flat (lever plane on the bed),
   6 walls, 50 % gyroid; hydro-dip or wrap for the carbon finish.
6. Slot lets the blade slide ±5 mm toward / away from the lever.



PLAN (lever plane)



REAR PLATE (outboard)



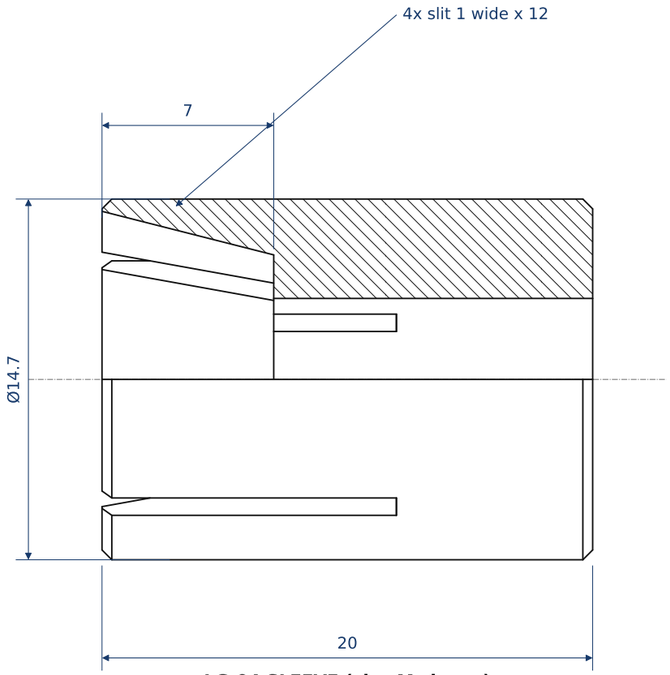
FRONT VIEW (outer face)

RACING LEVER GUARD / UNIVERSAL BAR-END FIT

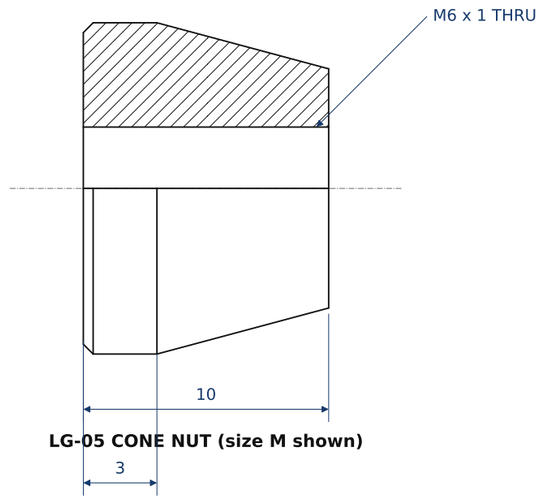
GUARD BLADE WITH BEND GUSSET

LG-03

MATERIAL: 3K 2x2 twill carbon / epoxy, forged-carbon gusset				FINISH: Gloss clear coat, edges R0.6	
SCALE: 3 : 2	UNITS: mm	QTY / SIDE: 1	GENERAL TOLERANCES: ISO 2768-mK	PROJECTION	SHEET: 4 / 6
SOURCE: cad/lever_guard.py (CadQuery), drawings.py				DATE: 2026-10-23	BREAK SHARP EDGES: 0.2 - 0.5



LG-04 SLEEVE (size M shown)



LG-05 CONE NUT (size M shown)

SIZES (all lengths as drawn: sleeve 20, recess 7, cone land 3)

Size	Bar ID	Sleeve OD	Recess Ø face / bottom	Cone Ø big / small	Taper (half)
S	13 - 15	12.7	11.7 / 8.4	11.5 / 8	14°
M	15 - 17	14.7	13.7 / 10.15	13.5 / 9.75	15°
L	17 - 19	16.7	15.7 / 12.15	15.5 / 11.75	15°

NOTES

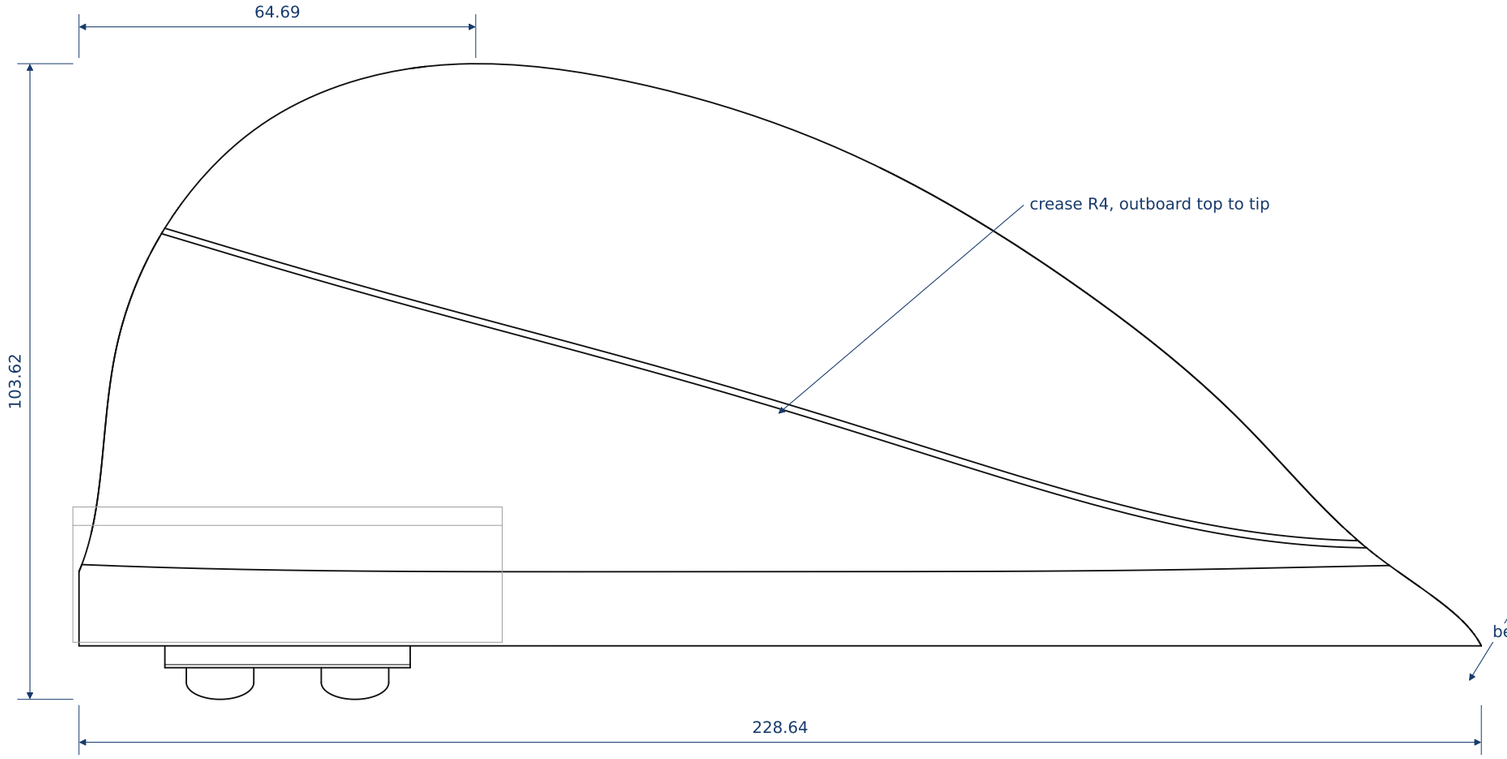
1. Bolt chain: M6 x 45 (ISO 4762) through LG-02 and LG-01 into the cone nut: 8.5 mm thread engagement at fit-up.
2. Tightening pulls the cone into the slit sleeve, which grips the bar bore. 8 N·m. Hold the arm square on its serration while tightening.
3. Pick the size from the bar's measured inside diameter.
4. Sleeve can be printed in TPU 95A (100 % infill) or cut from NBR tube; cone may be steel for higher load.

RACING LEVER GUARD / UNIVERSAL BAR-END FIT

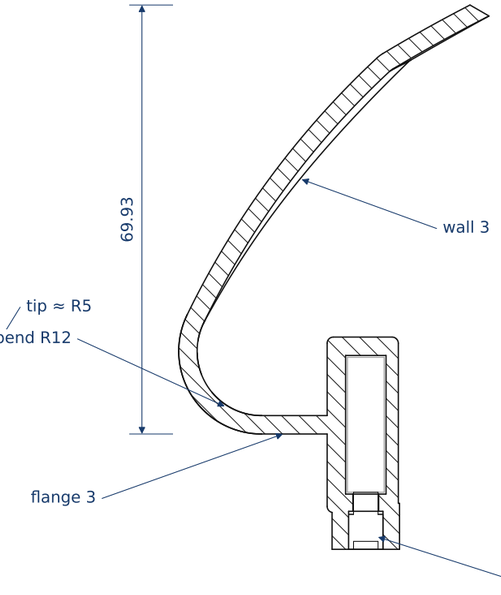
EXPANDER SET

LG-04 / LG-05

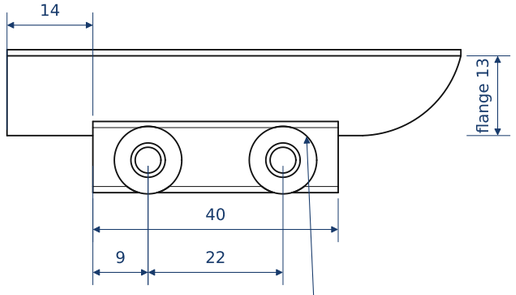
MATERIAL: LG-04 TPU 95A or NBR 70A; LG-05 6061-T6				FINISH: LG-05 clear anodise or zinc	
SCALE: 4 : 1	UNITS: mm	QTY / SIDE: 1 sleeve + 1 cone	GENERAL TOLERANCES: ISO 2768-mK	PROJECTION	SHEET: 5 / 6
SOURCE: cad/lever_guard.py (CadQuery), drawings.py				DATE: 2026-10-23	BREAK SHARP EDGES: 0.2 - 0.5



LG-06 SHELL (RH), SEEN FROM THE FRONT OF THE BIKE - blade phantom

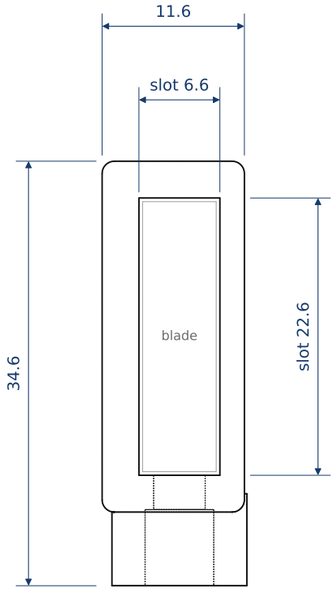


SECTION THROUGH A GRUB SCREW (1:1)



FLANGE AND SLOTTED MOUNT FROM BELOW

SLOTTED MOUNT, END VIEW (2:1)



NOTES

1. LG-06 surface as the STEP files 06 (RH) / 06b (LH, mirror image). One piece: shell, flange and slotted mount. Injection-moulded HDPE or PP, or printed ASA / PA12 with tree supports. Edges R1.
2. Press an M4 x 5.7 brass heat-set insert into each boss (soldering iron at about 230 °C), flush with the boss face.
3. Slide the mount onto the blade run from the blade tip, shell in front of the lever. The slot closes round the blade on all sides.
4. Lock with 2x M4 x 8 grub screws (ISO 4029, cup point, nylon-tipped if available) from below onto the blade's lower edge, 1 N·m.
5. Loosen to slide: 15-60 mm from the bend. No holes in the blade.
6. Keep the blade slot at -4 or more with the shell fitted (>= 10 mm to the reference lever). Check steering lock, mirrors, reservoir.

RACING LEVER GUARD / UNIVERSAL BAR-END FIT

WIND GUARD SHELL WITH SLOTTED MOUNT — LG-06

MATERIAL				FINISH	
HDPE or PP, moulded; ASA / PA12 printed				colour to order, edges R1	
SCALE	UNITS	QTY / SIDE	GENERAL TOLERANCES	PROJECTION	SHEET
1:1 / 2:1	mm	1 pair (RH + LH)	ISO 2768-mK		6 / 6
SOURCE				DATE	BREAK SHARP EDGES
cad/lever_guard.py (CadQuery), drawings.py				2026-10-23	0.2 - 0.5